Summary in Graph

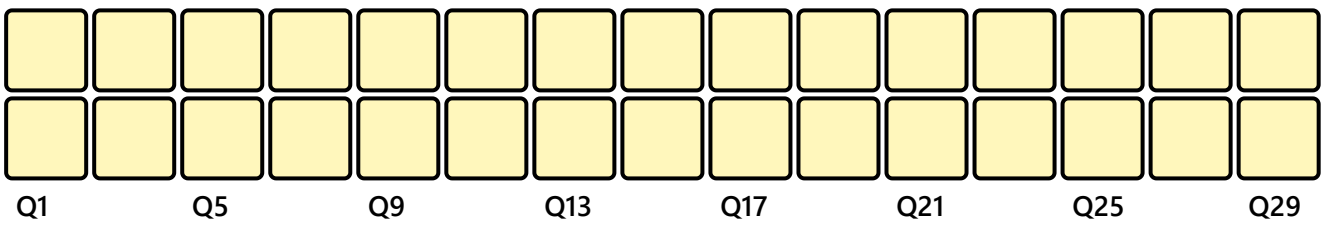
Exam Summary (GATE Overflow CS Test Series 2026 | Compiler Design | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (5) out of 47 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Compiler Design

When we merge states in an $LR(1)$ parser to form an $LALR(1)$ parser, we may introduce

A. shift-reduce conflict
B. reduce-reduce conflict
C. neither shift-reduce nor reduce-reduce conflict
D. both shift-reduce as well as reduce-reduce conflict

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Suppose we have a rightmost derivation which proceeds as follows:

$$\begin{aligned} S &\rightarrow Aabw \\ &\rightarrow ABw \end{aligned}$$

Which of the following is a possible handle for it?

A. $A \rightarrow ab$
B. $A \rightarrow a$
C. $S \rightarrow A$
D. $B \rightarrow ab$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Replacing the expression $19 * 7 \cdot 29$ by $133 \cdot 551$ is an example of _____

A. Strength reduction
B. Common sub-expression elimination
C. Constant folding
D. Constant propagation

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

Which of the following is/are correct?

A. L-attributed definitions can always be evaluated by the depth first visit of the parse tree
B. S-attributed definitions can always be evaluated by the depth first visit of the parse tree
C. Every L-attributed Syntax Directed Translation is S-attributed
D. Inherited attributes can always be evaluated by a classical preorder traversal

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

Which of the below relations is/are TRUE regarding GRAMMARS? (Mark all the appropriate choices)

A. $LL(1) \subset LALR(1) \subset LR(1)$
B. $SLR(1) \subset LALR(1) \subset LR(1)$
C. $\epsilon - \text{free } LL(1) \subset SLR(1) \subset LR(1)$
D. $\epsilon - \text{free } LL(1) \subset SLR(1) = LR(1)$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type Award: 1 Penalty: 0 Compiler Design

Which of the following is $TRUE$ regarding $LL(0)$ grammar? (Mark all the appropriate choices)

A. We can have an $LL(0)$ grammar for any finite language but not necessarily for any regular language
B. We can have an $LL(0)$ grammar for a finite language if and only if it does not contain empty string
C. We can have an $LL(0)$ grammar for any finite language if and only if it has prefix property
D. We cannot have an $LL(0)$ grammar for any regular language if it contains more than one string

Your Answer: Correct Answer: D Not Attempted Time taken: 00min 00sec Discuss

Q #7 Multiple Select Type Award: 1 Penalty: 0 Compiler Design

Which of the following is TRUE regarding the running time of an $LR(1)$ parser? (Mark all the appropriate choices)

A. It runs in linear time for all inputs
B. It runs in polynomial time for all inputs
C. For some inputs it may take exponential time
D. It runs in $O(n^3)$ but not always $O(n^2)$

Your Answer: Correct Answer: A;B Not Attempted Time taken: 00min 00sec Discuss

Q #8 Multiple Select Type Award: 1 Penalty: 0 Compiler Design

Which of the following statements regarding $LR(0)$ parser is/are FALSE? (Mark all the appropriate choices)

A. A $LR(0)$ configurating set cannot have multiple reduce items
B. A $LR(0)$ configurating set cannot have both shift as well as reduce items
C. If a reduce item is present in a $LR(0)$ configurating set it cannot have any other item
D. An $LR(0)$ parser can parse any regular grammar

Your Answer: Correct Answer: D Not Attempted Time taken: 00min 00sec Discuss

Q #9 Multiple Select Type Award: 1 Penalty: 0 Compiler Design

Which of the following sentences is/are $CORRECT$? (Mark all the appropriate choices)

A. A top-down parse produces a leftmost derivation of a sentence
B. A bottom-up parse produces a rightmost derivation of a sentence
C. A top-down parse produces a rightmost derivation of a sentence in reverse
D. A bottom-up parse produces a leftmost derivation of a sentence in reverse

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 01sec Discuss

Q #10 Multiple Select Type Award: 1 Penalty: 0 Compiler Design

Which of the following sentences regarding Viable prefixes is/are $CORRECT$? (Mark all the appropriate choices)

A. Viable prefixes are the set of prefixes of right-sentential forms that can appear on the stack of a shift-reduce parser
B. Viable prefixes are the set of prefixes of right-sentential forms that do not extend past the end of the right-most handle
C. Viable prefixes can be recognized using a Deterministic Finite Automata
D. Viable prefixes have prefix-free property (no string is a prefix of any other string)

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Which of the following languages are $LL(1)$ (an $LL(1)$ grammar $DOES$ exist for it)? (Mark all the appropriate choices)

A. $\{a^n o b^n \mid n \geq 1\} \cup \{a^n b^n \mid n \geq 1\}$
B. $\{a^n b^m \mid m \geq n\}$
C. $\{a^n b^m \mid n \geq m\}$
D. $\{a^n b^m \mid n = m\}$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Which of the following statements is/are FALSE?(Mark all the appropriate choices)

A. Any DCFL has an equivalent grammar that can be parsed by an $SLR(1)$ parser with an end string delimiter
B. Languages of grammars parsed by $LR(2)$ parsers is a strict super set of the languages of grammars parsed by $LR(1)$ parsers
C. Languages of grammars parsed by $LL(2)$ parsers is a strict super set of the languages of grammars parsed by $LL(1)$ parsers
D. There is no DCFL which is not having a grammar that can be parsed by an $LR(1)$ parse

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

The number of tokens in the following C code is _____

```c
/* GO2022 && GATE2023 */
int main()
{
    int x = 2022, y = 2023;
    printf("The sum is :%d",x+y);
    //The sum is:2022+2023
    return 0;
}
```

Your Answer: | Correct Answer: 27 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

The number of tokens in the following C code segment is _____

```
switch(value)
{
    case 1 : y = z*w; continue;
    case 2 : a = b***c; break;
    default : y = y++; break;
}
```

Your Answer: | Correct Answer: 39 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider the following grammar:

- $E \rightarrow TE'$
- $E' \rightarrow +TE' \mid \epsilon$
- $T \rightarrow FT'$
- $T' \rightarrow *FT' \mid \epsilon$
- $F \rightarrow (E) \mid id$

In the predictive parser table M, of the grammar, the entries $M[E, id]$ and $M[T', \$]$ respectively are _____

A. $E \rightarrow TE'$ and $T' \rightarrow *FT'$
B. $E \rightarrow TE'$ and $T' \rightarrow \epsilon$
C. $E' \rightarrow +TE'$ and $T' \rightarrow *FT'$
D. $E \rightarrow id$ and $T' \rightarrow \epsilon$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the grammar given below:

- $S \rightarrow ABA$
- $A \rightarrow Bc \mid dA \mid \epsilon$
- $B \rightarrow eA$

Let a, b, c and $\$$ be indexed as follows:

c	d	e	$\$$
2	3	5	7

If α is the index values for the symbols in the FOLLOW set of the non-terminal A, β is the index values for the symbols in the FOLLOW set of the non-terminal B and γ is the index values for the symbols in the FOLLOW set of the non-terminal S, all in their ascending orders, the value of $\alpha - \beta + \gamma =$ _____

(For example, if the FOLLOW set is $(c, d, e, \$)$, then the answer should be 2357)

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider the grammar:

- $S \rightarrow aABe$
- $A \rightarrow Abc \mid b$
- $B \rightarrow d$

For the sentence $abbcde$, the handles in the right-sentential form of the reduction are:

A. Abc, d and $aABe$
B. $aABe, d$ and b
C. Abc and $aABe$
D. $b, aABe, d$ and Abc

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Which of the following grammar is/are left-recursive? (Mark all the appropriate choices)

A. $A \rightarrow B \mid a \mid CBD$
$B \rightarrow C \mid b$
$C \rightarrow A \mid c$
$D \rightarrow d$
B. $A \rightarrow Ba \mid C$
$B \rightarrow AA$
$C \rightarrow B \mid b$
C. $Q \rightarrow QED \mid q$
$E \rightarrow e$
$D \rightarrow NFA \mid d$
$N \rightarrow DFA \mid n$
$F \rightarrow f$
$A \rightarrow a$
D. $S \rightarrow (L) \mid x$
$L \rightarrow SL'$
$L' \rightarrow \epsilon \mid , SL'$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

The attribute of three arithmetic operators in some programming language are given below.

OPERATOR	PRECEDENCE	ASSOCIATIVITY	ARITY
$+$	Low	Right	Binary
$-$	Low	Right	Binary
$*$	High	Left	Binary

The value of the expression $2 + 3 - 7 * 11$ in this language is _____

Your Answer: | Correct Answer: -72 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following grammar:

- $S \rightarrow AS \mid b$
- $A \rightarrow SA \mid a$

The above grammar is ____ (Mark all the appropriate choices)

A. $LL(1)$
B. $LR(0)$
C. $SLR(1)$
D. Ambiguous grammar

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following grammar:

$S \rightarrow Sa \mid a$

The above grammar (without any augmentation) is _____ (Mark all the appropriate choices)

A. $LL(1)$
B. $LR(0)$ and $SLR(1)$
C. $LALR(1)$
D. $LR(1)$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following grammar:

- $S \rightarrow aAd \mid bBd \mid aBe \mid bAe$
- $A \rightarrow c$
- $B \rightarrow c$

The above grammar is _____ (Mark all the appropriate choices)

A. $LL(1)$
B. $LR(0)$ and $SLR(1)$
C. $LALR(1)$
D. $LR(1)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following grammar: $S \rightarrow abS \mid acS \mid c$ The number of states in canonical set of $LR(0)$ items is _____

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following program.

```
int x;
int main() {
    x = 10;
    f();
    g();
}
void f() {
    int x = 20;
    h();
}
void g() {
    int x = 2;
    h();
}
void h() {
    printf("%d\n",x);
}
```

If α and β are the sum of all the values printed by the above code assuming static and dynamic scoping respectively, then $\alpha + \beta =$ _____

Your Answer: | Correct Answer: 42 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

A shift reduce parser carries out the actions specified within braces immediately after reducing with the corresponding rule of grammar:

$$\begin{aligned} S &\rightarrow S + T \quad \{\text{print } (\text{"+"})\} \\ &\quad |T \quad \{\ \} \\ T &\rightarrow T * F \quad \{\text{print } (\text{"*"})\} \\ &\quad |F \quad \{\ \} \\ F &\rightarrow \text{num} \quad \{\text{print } (\text{num.val})\} \end{aligned}$$

What is the translation of $5 + 7 * 3$ using the syntax-directed translation scheme described by the above rules?

A. 26
B. $573 * +$
C. $573 + *$
D. $+ * 573$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following Syntax Directed Translation Scheme, with non-terminals $\{S, T, F\}$ and terminals $\{\text{num}\}$. $S \rightarrow S\#T \quad \{S.val = S.val * T.val;\} \qquad |T \quad \{S.val = T.val;\}$
$T \rightarrow T\&F \quad \{T.val = T.val + F.val;\} \qquad |F \quad \{T.val = F.val;\}$
$F \rightarrow \text{num} \quad \{F.val = num.lval;\}$ The value of $2\#3\&1\#4\&3\#2\&5\#1\&6\#2$ using the syntax directed translation scheme described by the above rules is _____

Your Answer: | Correct Answer: 5488 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

The least number of total variables required to create a three-address code in static single assignment form for the expression $a + b * c/3 + d/e$ is _____

Your Answer: | Correct Answer: 10 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28

Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following code segment:

```
x = u + t;
y = u * v;
x = y + w;
y = x / z;
```

The minimum number of total variables required to convert the above code segment to static single assignment form is _____

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29

Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following intermediate program in three address code:

```
x = a * 2
y = x + a
x = y * x
y = y + 5
```

Which one of the following corresponds to a static single assignment form of the above code? (Mark all the appropriate choices)

A.

$x_1 = a * 2$
$y_1 = x + a$
$x_2 = y * x$
$y_2 = y + 5$

B.

$x_3 = a * 2$
$y_3 = x_3 + a$
$x_2 = y_3 * x_3$
$y_1 = y_3 + 5$

C.

$x_1 = a * 2$
$y_1 = x_1 + a$
$x_2 = y_1 * x_1$
$y_2 = y_1 + 5$

D.

$x_1 = a * 2$
$y_1 = x_1 + a$
$x_2 = y_1 * x_1$
$y_1 = y_1 + 5$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30

Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

The program below uses three input variables a, b and c and two temporary variables d and e.

```
    d = a + b;
    e = c + d;
    d = c + a;
    e = d + b;
5.  return d * e;
```

Assuming that all operations take their operands from registers, the minimum number of registers needed to execute this program without spilling is _____

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

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